Natural Language Processing (NLP)

CountVectorizer

--> To convert text into number

In [1]:
```
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
```

In [2]:
```
text = ['Hello my name is james','james this is my python notebook','james try
```

In [3]:
```
count_vect = CountVectorizer()
```

In [4]:
```
count_matrix = count_vect.fit_transform(text)
```

In [5]:
```
count_array = count_matrix.toarray()
```

In [6]:
```
df = pd.DataFrame(data=count_array,columns = count_vect.get_feature_names_out(
```

In [7]:
```
df
```

Out[7]:

	big	count	create	dataset	differnt	features	hello	is	james	my	name	r
0	0	0	0	0	0	0	1	1	1	1	1	
1	0	0	0	0	0	0	0	1	1	1	0	
2	1	0	1	1	0	0	0	0	1	0	0	
3	0	0	0	0	1	0	0	0	1	0	0	
4	0	1	0	0	0	1	0	0	0	0	0	

Parameter

- Lowercase

--> Convert all characters to lowercase before tokenizing . Default is set to true and takes boolean value.

In [8]:
```
text = ['hello my name is james','Hello my name is James']
```

In [10]:
```
count_vect = CountVectorizer(lowercase=False)
```

In [11]:
```
count_matrix = count_vect.fit_transform(text)
```

In [12]:
```
count_array = count_matrix.toarray()
```

In [13]:
```
df = pd.DataFrame(data=count_array,columns = count_vect.get_feature_names_out(
```

In [14]:
```
df
```

Out[14]:

	Hello	James	hello	is	james	my	name
0	0	0	1	1	1	1	1
1	1	1	0	1	0	1	1

- Stop_words

In [15]:
```
text = ['Hello my name is james','james this is my python notebook','james try
```

In [16]:
```
count_vect = CountVectorizer(stop_words= ['is','to','my'])
count_matrix = count_vect.fit_transform(text)
```

In [17]:
```
count_array = count_matrix.toarray()
```

In [18]:
```
df = pd.DataFrame(data=count_array,columns = count_vect.get_feature_names_out(
```

In [19]:
```
df
```

Out[19]:

	big	count	create	dataset	differnt	features	hello	james	name	notebook
0	0	0	0	0	0	0	1	1	1	(
1	0	0	0	0	0	0	0	1	0	1
2	1	0	1	1	0	0	0	1	0	(
3	0	0	0	0	1	0	0	1	0	(
4	0	1	0	0	0	1	0	0	0	(

- stop_words = Whole english language (helping verb like 370 words)

In [20]:
```
count_vect = CountVectorizer(stop_words= 'english')
count_matrix = count_vect.fit_transform(text)
count_array = count_matrix.toarray()
df = pd.DataFrame(data=count_array,columns = count_vect.get_feature_names_out(
df
```

Out[20]:

	big	count	create	dataset	differnt	features	hello	james	notebook	pytho
0	0	0	0	0	0	0	1	1	0	
1	0	0	0	0	0	0	0	1	1	
2	1	0	1	1	0	0	0	1	0	
3	0	0	0	0	1	0	0	1	0	
4	0	1	0	0	0	1	0	0	0	

Using max_df and min_df(covered later)

- Max_df:

In [24]:

```
count_vect = CountVectorizer(max_df=1)
count_matrix = count_vect.fit_transform(text)
count_array = count_matrix.toarray()
df = pd.DataFrame(data=count_array,columns = count_vect.get_feature_names_out(
df
```

Out[24]:

	big	count	create	dataset	differnt	features	hello	name	notebook	pytho
0	0	0	0	0	0	0	1	1	0	
1	0	0	0	0	0	0	0	0	1	
2	1	0	1	1	0	0	0	0	0	
3	0	0	0	0	1	0	0	0	0	
4	0	1	0	0	0	1	0	0	0	

In [27]:

```
count_vect = CountVectorizer(max_df=0.75)
count_matrix = count_vect.fit_transform(text)
count_array = count_matrix.toarray()
df = pd.DataFrame(data=count_array,columns = count_vect.get_feature_names_out(
df
# Here 0.75 is 75% --> occuring the word
```

Out[27]:

	big	count	create	dataset	differnt	features	hello	is	my	name	noteboo
0	0	0	0	0	0	0	1	1	1	1	
1	0	0	0	0	0	0	0	1	1	0	
2	1	0	1	1	0	0	0	0	0	0	
3	0	0	0	0	1	0	0	0	0	0	
4	0	1	0	0	0	1	0	0	0	0	

- Min_df:

In [29]:
```
count_vect = CountVectorizer(min_df =2)
count_matrix = count_vect.fit_transform(text)
count_array = count_matrix.toarray()
df = pd.DataFrame(data=count_array,columns = count_vect.get_feature_names_out(
df
```

Out[29]:

	is	james	my	of	to
0	1	1	1	0	0
1	1	1	1	0	0
2	0	1	0	0	1
3	0	1	0	1	1
4	0	0	0	1	0

In [ ]: